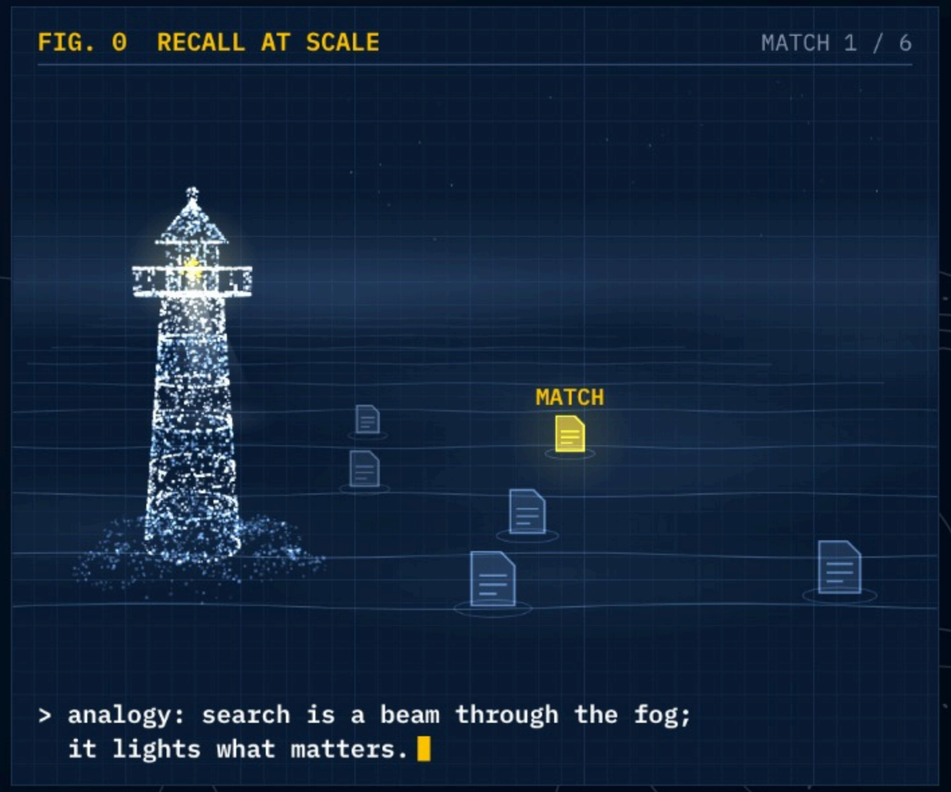
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THE MEMORY LAYER · PART 4

Recall at scale

Keyword, meaning and hybrid search for a brain that has outgrown its index. Measured, not guessed.



10–minute video | All roles | Runs locally

ALEX

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TODAY

By the end, your learners can…

1

Explain when index.md stops being enough, and why loading everything fails

2

Compare keyword (BM25) and meaning (embeddings) search, and when each wins

3

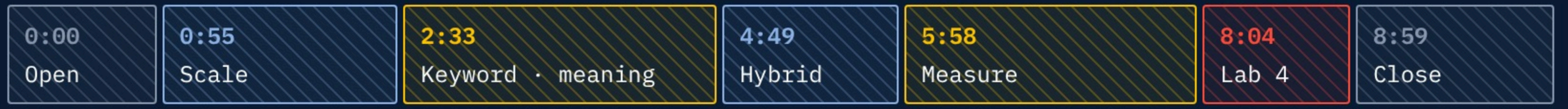
Run hybrid search with rank fusion over stamped, heading-aware chunks

4

Measure recall with a small answer key, and fix a miss

CHAPTERS · 10-MIN VIDEO

drawn to scale



JOSH

BLOCK 1 · WHY THIS MATTERS

Past a few hundred pages, search.

ANALOGY

> A corner shop vs. a warehouse: you can scan every shelf in one, not the other.

index.md works at small scale
About 100 sources, hundreds of pages (Karpathy)

Then it's too long to read
Long inputs bury the middle and burn tokens

Search returns the few that matter
brain search ranks pages; read only those



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BLOCK 1 · WHAT THE RESEARCH SAYS

More context isn’t better recall

53.8%

Answer buried mid-context

Below the 56.1% the model scored with no documents at all. Lost in the Middle, GPT-3.5

18/18

Models got worse

Every model tested degraded as the input grew, even on simple tasks. Chroma, 2025

−49%

Failed retrievals

Contextual embeddings plus keyword search; −67% with a reranker. Anthropic, 2024

+18%

Hybrid vs. keyword

Fusing keyword and meaning ranks beat BM25 alone on nDCG@10. Elastic (ELSER), 2023

Retrieve the **few pages that matter**. Combine **words** and **meaning**. Under about 200k tokens, just load it all.

ALEX

BLOCK 2 · KEYWORD SEARCH

Keyword search: rare words count most

ANALOGY

> A back-of-book index: look up “ORA-01017” and it sends you to page 212.

Rare terms weigh most

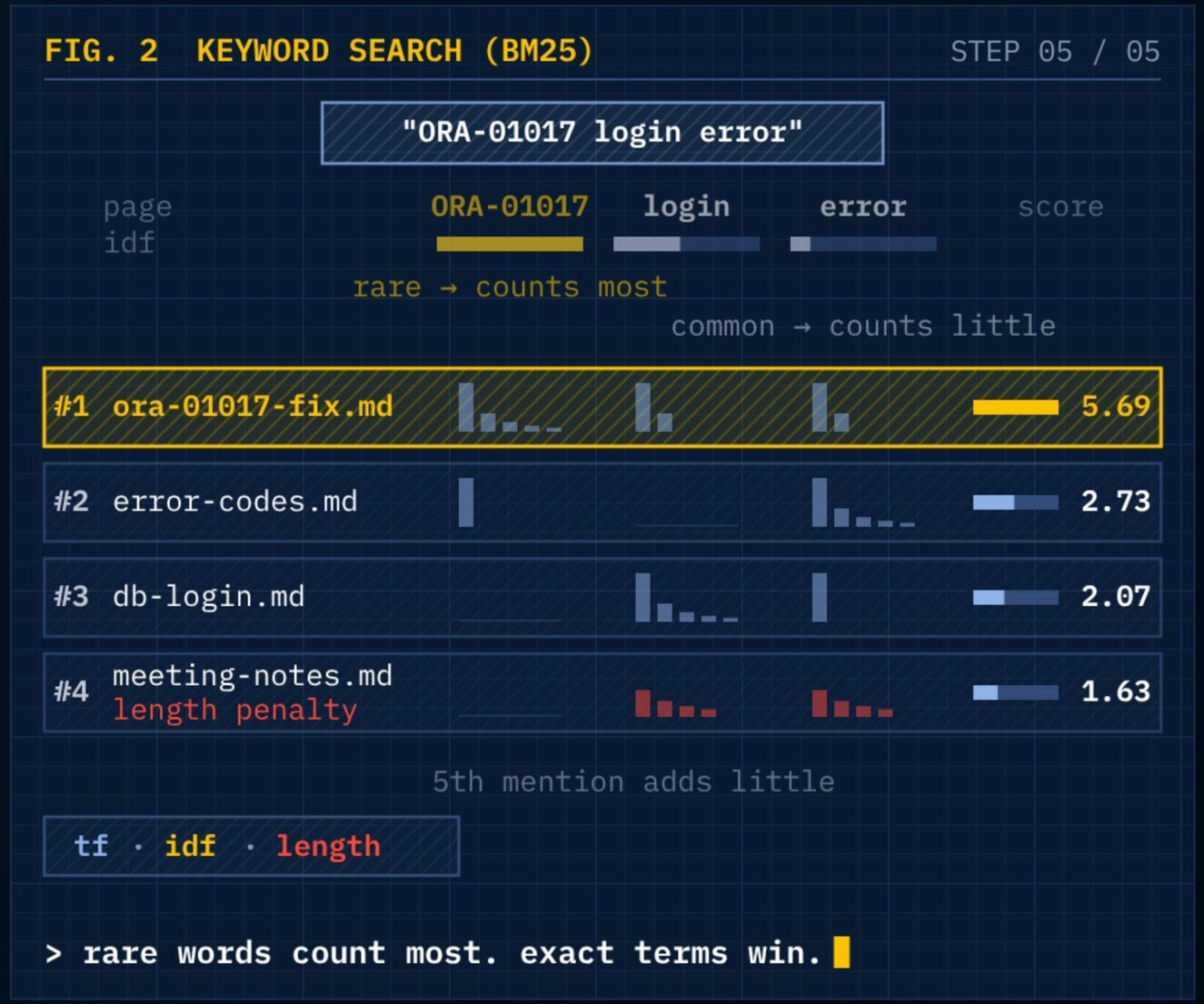
An error code beats a word like “error” (IDF)

Repeats help, less and less

The 5th mention adds little (k1)

Long pages can’t cheat

Length normalization (b = 0.75)



BLOCK 2 · SEARCH BY MEANING

Search by meaning, not words

ANALOGY

> A librarian who knows meaning: ask about “fixing my car” and she brings the vehicle binder.

Text becomes 768 numbers

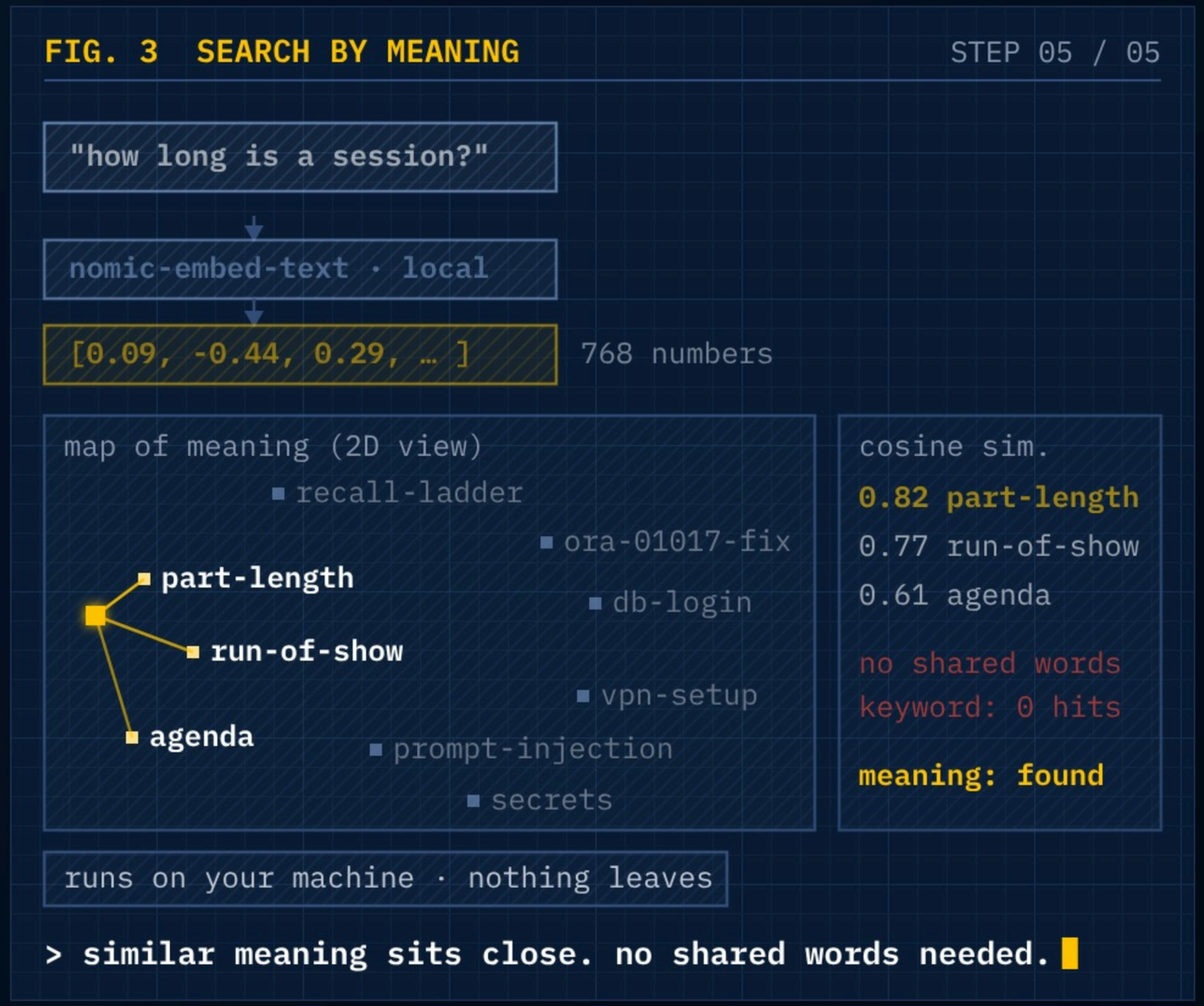
Close meanings land close together

Finds paraphrases

No shared words needed

Weak on exact tokens

Codes, IDs, numbers and “not”



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BLOCK 2 · LOCAL EMBEDDINGS

Keep the meaning on your machine

> brain index embeds each chunk once, and re-embeds only files that changed.

nomic-embed-text via Ollama

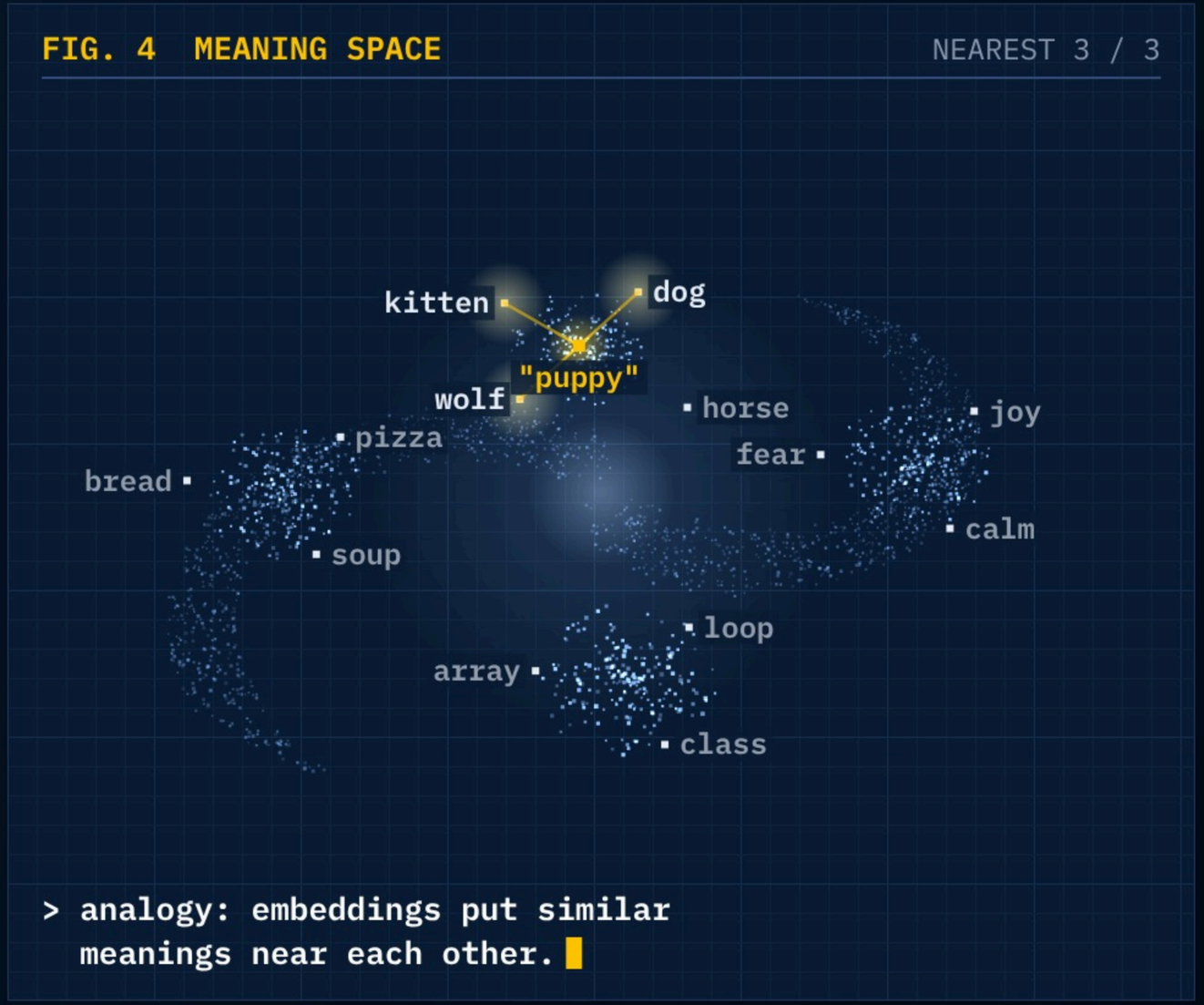
137M parameters, 768 dims, 274 MB, Apache-2.0

Nothing leaves the laptop

Notes and vectors stay in .brain/index.sqlite

No Ollama? Still works

brain search falls back to keyword



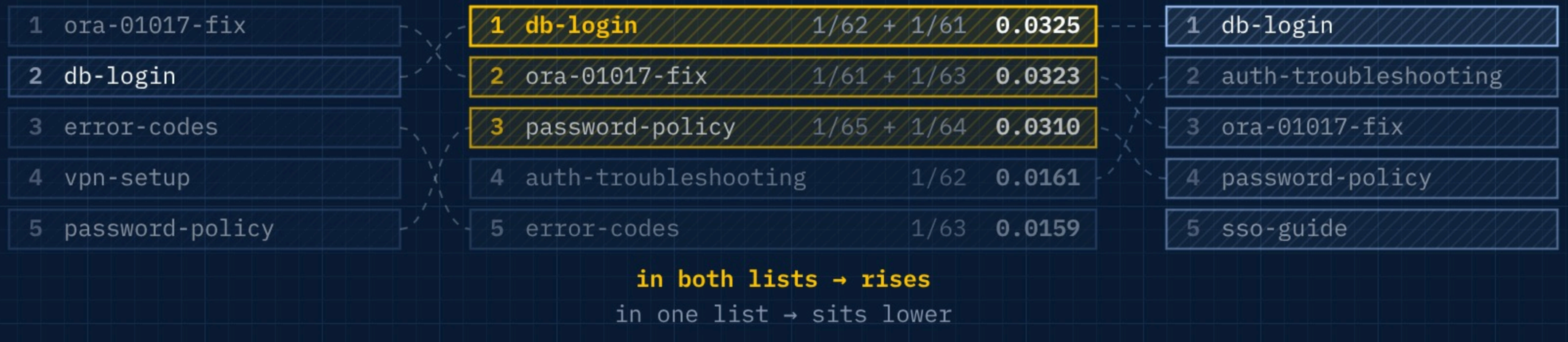
JOSH

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BLOCK 3 · HYBRID SEARCH

Hybrid: two judges, one ranking

FIG. 5 TWO JUDGES, ONE RANKING — STEP 04 / 04



> agreement rises. no scor

ANALOGY Different scales, so don't average: reward what **both** rank high. Each list adds **1 / (60 + rank)**.

JOSH

BLOCK 3 · CHUNKING

Cut at headings. Stamp every card.

ANALOGY

> Recipe cards with good titles and nicknames: one dish per card, easy to find.

Heading-aware chunks

Up to 220 words of text (≈290 tokens), split at headings

Stamp each chunk

The note's summary and aliases ride along

Aliases catch nicknames

Add the words your team actually uses



BLOCK 4 · MEASURE RECALL

Measure recall before you tune

ANALOGY

> A quiz with an answer key: retake the same quiz after every change.

A golden set

20–50 real questions, each with its right page

recall@5 and MRR

Was it found? How high was it?

Re-run after every change

Aliases, chunking, k1 and b, or a new model

FIG. 7 A QUIZ WITH AN ANSWER KEY — STEP 05 / 05

reference vault · 6 questions · rank of the right page

question → expected	KW	MEANING	HYBRID
streamable HTTP vs stdio → mcp-transports…	1	1	1
cosign → image-signing…	1	1	1
stop a web page hijack… → prompt-injection	2	4	1
why does it forget… → agent-memory-layers	5	2	3
which model if the vend… → ai-resilience	4	1	1
crawl walk run → five-stage-training…	1	3	1
recall@5	6/6	6/6	6/6
MRR	0.66	0.68	0.89

+ alias: crawl walk run

MRR = average of 1/rank

> measure before and after every change.

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BLOCK 4 · WHAT CAN GO WRONG

Three ways search fails quietly

ONE-SIDED

Hybrid buries a lone hit

A page only one judge finds can rank below pages both half-like. Check each mode.

STALE

The index is out of date

Search reads .brain/index.sqlite. Run brain index after edits and ingests.

SWITCHED

New model, old vectors

Vectors from different models don't mix. New model: re-index before you search.

The guardrails: golden set · re-index · check each mode · re-embed

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WALKTHROUGH

Same question, three searches

Watch

1	**Index the vault**	brain index --no-embed, then brain index with Ollama running.
2	**Three modes**	brain search "cosign" --mode keyword | semantic | hybrid
3	**A paraphrase**	“why does the assistant forget everything between chats?”
4	**Fix a miss**	Alias “crawl walk run”, brain index, run the quiz again.

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THE RULE OF THUMB

Words for **exact**. Meaning for ideas. Fuse the **ranks**. Measure every **change**.

When search can’t find a page, fix the page: its summary and aliases. Then retake the quiz.

ALEX

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HANDS-ON LAB 4

Index, score, fix.

Pause & try

A	**Index**	brain index --no-embed, then with Ollama. Check brain stats.
B	**Score**	python3 run_eval.py on golden.csv. Record ranks in 3 modes.
C	**Explain**	Pick a row where the modes disagree. Why?
D	**Fix**	Alias “crawl walk run”, brain index, score again.
E	**Extend**	Add one real question of your own to golden.csv.

ALEX

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CHECK FOR UNDERSTANDING

Can every learner answer these?

1

Why isn't index.md enough past a few hundred pages? Name one research result.

2

What three things does BM25 reward or penalize?

3

Give one query where keyword beats meaning, and one where meaning wins.

4

How does RRF merge two lists, and why use ranks instead of scores?

5

Why stamp each chunk with its note's summary and aliases?

6

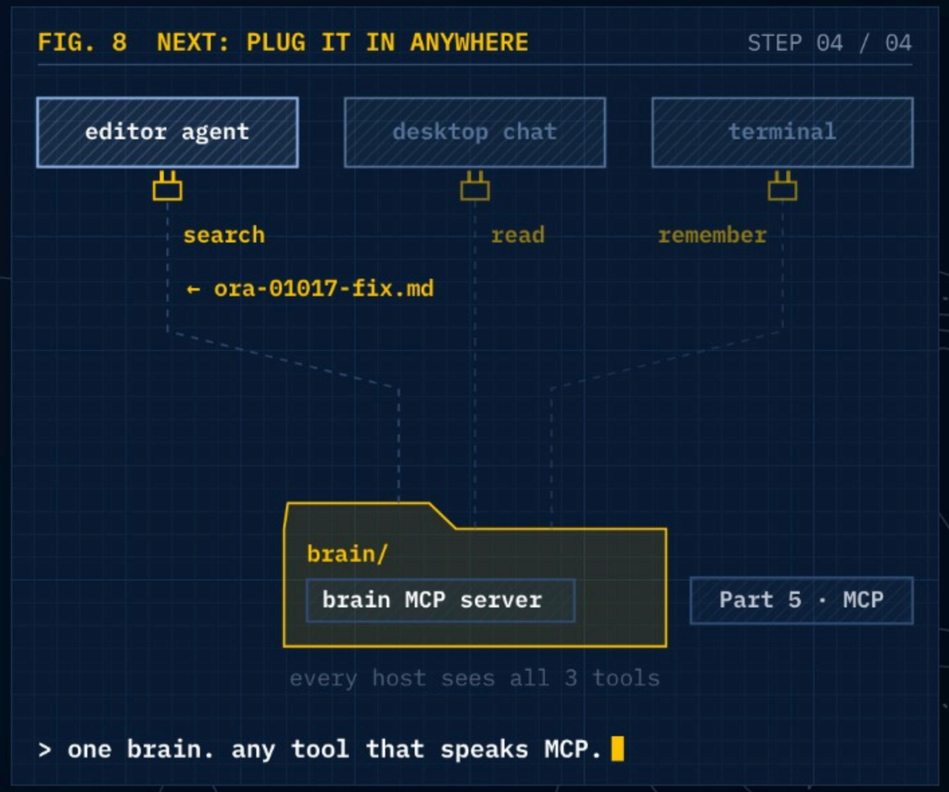
What do recall@5 and MRR tell you, and when do you re-run them?

JOSH

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NEXT · PART 5 · PLUG IT IN ANYWHERE

One brain, every tool that speaks MCP



What Part 5 covers

- The brain MCP server
- stdio vs. streamable HTTP
- Write guardrails
- The Obsidian CLI

ALEX